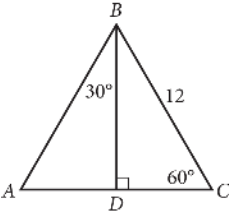


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

Answer

- A. 4
- B. 6
- C. $6\sqrt{2}$
- D. $6\sqrt{3}$

Correct Answer: B

Rationale

Choice B is correct. Triangles ADB and CDB are both $30^\circ - 60^\circ - 90^\circ$ triangles and share $\overline{BD}$. Therefore, triangles ADB and CDB are congruent by the angle-side-angle postulate. Using the properties of $30^\circ - 60^\circ - 90^\circ$ triangles, the length of $\overline{AD}$ is half the length of hypotenuse $\overline{AB}$. Since the triangles are congruent, $AB = BC = 12$. So the length of $\overline{AD}$ is $\frac{12}{2} = 6$.

Alternate approach: Since angle CBD has a measure of 30° , angle ABC must have a measure of 60° . It follows that triangle ABC is equilateral, so side AC also has length 12. It also follows that the altitude BD is also a median, and therefore the length of AD is half of the length of AC, which is 6.

Choice A is incorrect. If the length of $\overline{AD}$ were 4, then the length of $\overline{AB}$ would be 8. However, this is incorrect because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12. Choices C and D are also incorrect. Following the same procedures as used to test choice A gives $\overline{AB}$ a length of $12\sqrt{2}$ for choice C and $12\sqrt{3}$ for choice D. However, these results cannot be true because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12.